PREFACE

The fusion of cutting-edge technology and traditional methods has created new opportunities to improve wellness and health. With thousands of years of history, yoga is a discipline that has gained worldwide recognition for its comprehensive health benefits for both the body and the mind. However, the efficiency of yoga mostly depends on how precisely its poses are performed, and this accuracy is typically guaranteed by one-on-one training from skilled instructors. But as the digital era approaches, creative ways to duplicate the advantages of faceto-face coaching in a remote, automated setting are needed.

Using the most recent developments in deep learning and computer vision, this paper explores the creation of an advanced yoga pose identification programme. Through the integration of OpenCV for image processing and MediaPipe for pose landmark identification, this project seeks to offer a real-time feedback mechanism that guarantees users execute yoga poses correctly and safely. The Streamlit framework, upon which the product is based, provides administrators and end users with an easy-to-use interface and a role-based login mechanism. Administrators have the ability to enhance the pose database and offer comprehensive feedback, while end users can obtain prompt and accurate corrections to enhance their skills.

With the global COVID-19 pandemic adding to the already enormous demand for remote wellness options, our project aims to make high-quality yoga instruction more accessible to everybody. Its position at the nexus of tradition and innovation guarantees that, regardless of the practitioners' location, the age-old wisdom of yoga can be applied with contemporary accuracy. This study describes the technological process used to create a reliable, user-friendly, and efficient yoga posture identification system, from conception to implementation. It includes information on techniques, technologies, and results.

In addition to improving individual yoga practices, our project aims to stimulate more technological integration in the service of health and wellbeing. Through an in-depth explanation of our methodology and results, this paper hopes to make a significant contribution to the domains of artificial intelligence, computer vision, and digital health.